



Review

Effects of ambient ozone concentrations with different averaging times on asthma exacerbations: A meta-analysis



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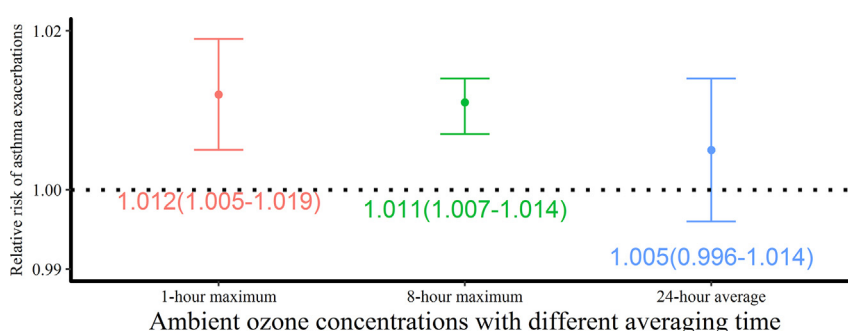
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HIGHLIGHTS

- Ozone measured as 1h or 8h daily maximum were more consistently associated with asthma exacerbations than 24h average.
- The association was significant during the warm season; and it's more robust in regions where ozone concentrations were higher.
- Our findings support the use of shorter averaging time of ozone within a day as a more biologically relevant measure.

GRAPHICAL ABSTRACT



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ABSTRACT

Background: Mounting evidence suggests that short-term exposure to ozone increases the risk of asthma exacerbations. However, ozone exposures have been assessed using ambient ozone concentrations averaged over different time periods in different studies.

Objective: To evaluate the risks for asthma exacerbations related to ambient ozone measured as 1-hour or 8-hour daily maximum and 24-hour average concentrations.

Methods: Based on a literature search in PubMed, EMBASE and Web of Science, we identified all time-series studies as of December 4th, 2018 and included 47 eligible studies in our analyses. Asthma exacerbation is defined as the risk for emergency room visits or hospital admissions. Pooled relative risks (RRs) and 95% confidence intervals (95% CIs) for a 10 µg/m³ increase in daily ozone concentration were estimated using random effect models. Subgroup analyses and sensitivity analyses were also performed to examine the risks for different seasons, regions and age groups and for the robustness of our main findings.

Results: Significant and similar associations were found for O₃–1 h max (RR, 1.012; 95%CI, 1.005–1.019) and O₃–8 h max (RR, 1.011; 95%CI, 1.007–1.014), while marginal effect was identified for O₃–24 h average (RR, 1.005; 95%CI, 0.996–1.014). No significant publication bias but high heterogeneities were observed. During the warm

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season, ozone was significantly associated with asthma exacerbation. O₃-1 h max had the highest RR of 1.014 (95%CI, 1.005–1.024), followed by O₃-8 h max (RR, 1.012; 95%CI, 1.009–1.016), while marginal association was identified for O₃-24 h avg (RR, 1.008; 95%CI, 0.998–1.017). During the cold season, null associations were identified for all the three averaging times. Variations were also observed in region and age.

Conclusion: Ozone exposure measured as 1-hour or 8-hour daily max were more consistently associated with asthma exacerbations than 24-hour average exposure during the warm season.

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1. Introduction

Asthma is one of the most prevalent chronic respiratory diseases, affecting an estimated 272.68 million people worldwide in 2017 (James et al., 2018). One of the principal strategies of asthma control and management is to prevent asthma exacerbations (Bostantzoglou et al., 2015) that are responsible for millions of emergency room visits (ERVs) and hospital admissions (HAs) each year (Jackson et al., 2011). To do so, identifying risk factors is an important first step. Ground-level ozone pollution is a well-known risk factor for asthma exacerbations. Anenberg et al. estimated that 9–23 million annual asthma ERVs could be attributable to ozone exposure globally in 2015 (Anenberg et al., 2018). Such estimates of ozone effects, however, are recognized to have large uncertainties, partly due to inconsistencies in exposure-response relationships.

The relationship between short-term exposure to ambient ozone and asthma exacerbations has been demonstrated in a large number of time-series studies (Cirera et al., 2012; Dai et al., 2018; Gharibi et al., 2018; Hajat et al., 1999; Havet et al., 2018; Hernández-Cadena et al., 2000; Ito et al., 2007; Jaffe et al., 2003; Ko et al., 2007; Lee et al., 2006a; Lee et al., 2002; Lee et al., 2006b; Lin et al., 2004; Mar and Koenig, 2009; Medina et al., 1997; Mohr et al., 2008; Morgan et al., 1998; Neidell and Kinney, 2010; Petroschevsky et al., 2001; Samoli et al., 2011; Schouten et al., 1996; Sheppard et al., 1999; Silverman and Ito, 2010; Son et al., 2013; Stieb et al., 1996; Stieb et al., 2009; Stroosnider et al., 2019; Sunyer et al., 1997; Szyzkowicz, 2008; Tenías et al., 1998; Thompson et al., 2001; Wilson et al., 2005; Zhang et al., 2018; Zu et al., 2017). However, different studies may have used different ozone exposure averaging times including 1-hour daily maximum (O₃-1 h max), 8-hour daily maximum (O₃-8 h max) and 24-hour daily average concentration (O₃-24 h avg). The selection of averaging time for ozone exposure, reflecting the availability of data or investigators' understanding of biological relevance of exposure time frame, is expected to affect the risk estimates or exposure-response relationships (Darrow et al., 2011; Sun et al., 2018). Ozone exposure using averaging time that reflect peak pollution levels (e.g., O₃-1 h max or O₃-8 h max)

may be more biologically relevant than 24-hr avg when the health effect is triggered by a high does spike (Darrow et al., 2011). On the contrary, O₃-24 h avg may be more representative of average exposures.

In the latest ozone National Ambient Air Quality Standards (NAAQS) review (EPA, 2015) that the US EPA used to quantify health effects, 37 epidemiologic studies were identified. Among those, 18 studies used O₃-1 h max, 15 used O₃-8 h max, and 14 used O₃-24 h avg. Several studies have investigated which ozone averaging time was more strongly associated with human health response. Sun et al. (2018) found that the acute mortality effect of ozone was more closely related to O₃-24 h avg than O₃-1 h max and O₃-8 h max. In contrast, a panel study found that O₃-8 h max was more closely associated with respiratory health variables (Li et al., 2018). To our knowledge, there have been no systematic evaluations to compare risk estimates for asthma exacerbations when different averaging times are used to represent short-term ozone exposure. The goal of the present meta-analysis is to evaluate and compare the risks for asthma exacerbation associated with ozone exposure estimated using three averaging times for ambient ozone concentrations (O₃-1 h max, O₃-8 h max and O₃-24 h avg).

2. Methods

This study complies with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (Moher et al., 2009).

2.1. Selection criteria and identifying studies

The PRISMA flow diagram is presented in Fig. 1.

The following inclusion criteria had to be fulfilled: (1) Studies focusing on associations between short-term ozone exposure and acute asthma exacerbations (with the lag duration no longer than 21 days); (2) Time-series study; (3) Studies in which one or more of the 3 averaging times were used.

We initially conducted searches identifying all published literature and meeting abstracts in EMBASE, PubMed, Cochrane Database of

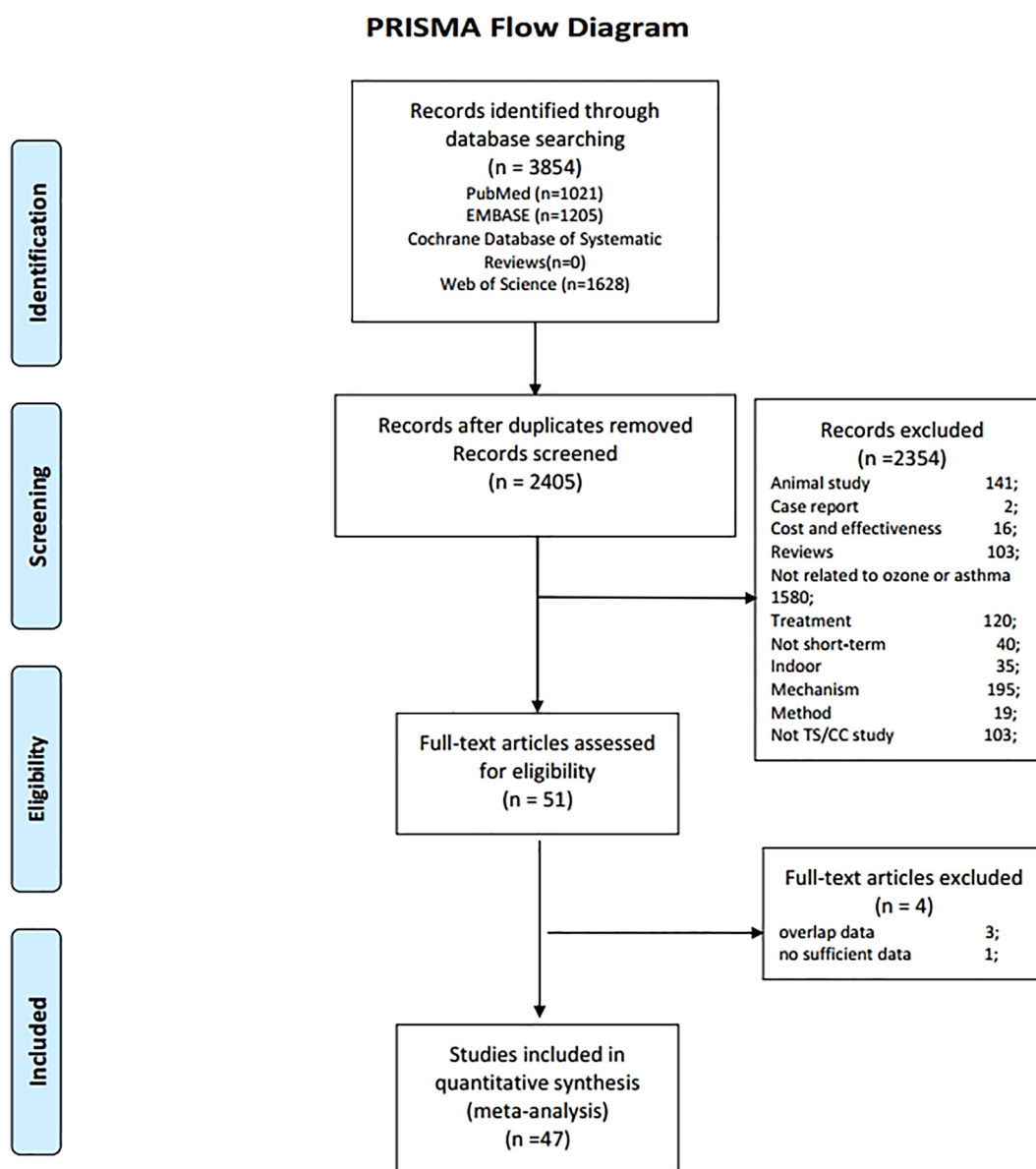


Fig. 1. Flow diagram of this meta-analysis.

Systematic Reviews and Web of Science without limitation to language. Articles were searched for the period up to December 4th, 2018 by using combinations of the following search terms, “ozone”, “O₃”, “asthma”, details of search strategies are presented in Supplementary Table 1. There are multiple expressions of averaging time for ozone exposure, short-term ozone exposure (*e.g.* acute, short-term, short term), study type (*e.g.* time-series, time series) and asthma exacerbations (*e.g.* hospitalizations, HA, EMV, emergency outpatient). In addition, the references of all identified publications, including previous meta-analyses, review articles were hand-searched for other relevant publications.

Citations were merged together in Endnote (version X7), which contributed to the retrieval of references. Duplicates were identified, inducing 2405 inclusions. Screening of these 2405 references led to 2354 exclusions. A thorough review of the 51 remaining articles included the following efforts. In case of overlapping samples, we examined the most recently described and/or the largest sample. Studies with missing data were excluded if no reply was obtained despite repeated contacts with corresponding authors. Finally, 47 full-text articles were included in the present meta-analysis.

2.2. Data collection

For each of the selected studies, the following data elements were extracted including, author, year of publication, relative risks (RRs) and 95% confidence intervals (95%CI), country, study period, season, age group, asthma exacerbation measures, asthma diagnosis standard, the quality of measurement methods and adjustments (long-term trend, seasonality, temperature, humidity, days of the week, holidays and influenza epidemics), the choice of averaging time for ozone exposure, season, lag pattern used, ozone concentration and daily average temperature during the study period. In the case of multicity studies, a city-specific RR was extracted when it was available.

2.3. Statistical analysis

To allow for comparability among multiple ozone averaging times, all estimates should be based on the same measure of ozone concentration such as O₃-8 h max. We converted all estimates to O₃-8 h max through assumption of a standard ratio 20:15:8 for O₃-1 h max: O₃-

8 h max: O₃-24 h avg (Schwartz, 1997). This is a ratio widely used to convert among ozone averaging times (Bell et al., 2005; EPA, 2013; Wong et al., 2008; Yin et al., 2017). For example, a 10 µg/m³ increase in the O₃-1 h max concentration corresponds to a 7.5 µg/m³ increase in the daily O₃-8 h max concentration. And estimates presented in parts per billion (ppb) were converted to µg/m³ using 1.96 µg/m³ per 1 ppb for the standard temperature and pressure (Thurston and Ito, 2001; Yin et al., 2017). Results reported in various forms were converted to regression coefficients and standard errors for pooling (Bell et al., 2014). The overall effect estimates reported in this study were standardized as RRs (95%CI) per 10 µg/m³ increase of O₃-8 h max.

There was lag effects (measured in days) between short-term ozone exposure and asthma exacerbations; however, included studies varied in lag selection patterns. Given the lack of standardized methods of reporting, we adopted *a priori* lag selection scheme proposed by Atkinson et al. (Cho et al., 2019). If only one lag estimate was reported, it was directly included for analyses. If multiple lag estimates were reported, we selected effects for one lag based on (1) a single-day lag was preferred rather than cumulative lag; and (2) the most frequently used lag was used.

A random-effects model was used to calculate the pooled RRs (Song et al., 2001). Between-study heterogeneity and between-study inconsistency were assessed by using Cochrane Q statistic and by estimating *I*², respectively (Higgins et al., 2003). Evidence of publication bias was determined using funnel plot and Egger's statistical test (Egger et al., 1997). A meta-regression was performed to explore the influence of daily average temperature on the associations between ozone and asthma exacerbations.

To evaluate the robustness of meta-analysis results, we conducted sensitivity analyses. First, we assessed the quality of original studies to examine if quality of study methodology would influence the results. Since there has been no established quality assessment tool for time-series studies (Achilleos et al., 2017), we adapted the quality assessment tool based on Aris et al.'s study (Aris et al., 1993), which was also applied in our previous study (Zheng et al., 2015). Items assessed include the validation of asthma diagnosis (0 to 1 point), the quality of ozone exposure measures (0 to 1 point), and the extent of adjustment for confounders (0 to 3 points). Asthma exacerbation was confirmed if it was coded by International Classification of Diseases (ICD) (0 for no valid criteria). The frequency of measurement and missing data were considered (1 point for measurements performed daily with <25% missing data, otherwise 0 point was assigned). Regarding the potential confounders, 1 point was scored if long-term trends, seasonality and temperature were all adjusted, otherwise 0 point was assigned. Each additional adjustment was added for 1 point (the total point for confounder adjustment is 3 at most). Based on the quality assessment, we restricted the analyses to studies with high quality (quality score of 5) to examine if study quality would influence the results. Second, we restricted the analyses to studies with lag days ≤ 3 to avoid confounding of lag day patterns. Third, we presented the original overall RRs and 95% CIs, before converting a different time-averaged ozone concentration to O₃-8 h max, to see if the risk estimates among averaging times were robust before and after the 8-hour based standardization.

Statistical analyses were conducted using R (version 3.5.0). All tests were two-sided and statistical significance was defined as *P* < 0.05.

3. Results

Of the 47 included studies (Amancio and Nascimento, 2012; Andersen et al., 2008; Anderson et al., 1998; Babin et al., 2008; Cassino et al., 1999; Castellsague et al., 1995; Chakraborty et al., 2014; Chen et al., 2016; Cirera et al., 2012; Dai et al., 2018; Fusco et al., 2001; Galan et al., 2003; Gharibi et al., 2018; Gouveia and Fletcher, 2000; Hajat et al., 1999; Havet et al., 2018; Hernández-Cadena et al., 2000; Ito et al., 2007; Jaffe et al., 2003; Ko et al., 2007; Kuo et al., 2018; Lee

et al., 2006a; Lee et al., 2002; Lee et al., 2006b; Lin et al., 2004; Mar and Koenig, 2009; Medina et al., 1997; Mohr et al., 2008; Morgan et al., 1998; Neidell and Kinney, 2010; Petroeschovsky et al., 2001; Samoli et al., 2011; Schouten et al., 1996; Sheppard et al., 1999; Silverman and Ito, 2010; Son et al., 2013; Stieb et al., 1996; Stieb et al., 2009; Strosnider et al., 2019; Sunyer et al., 1997; Szyszkowicz, 2008; Tenias et al., 1998; Tetreault et al., 2016; Thompson et al., 2001; Wilson et al., 2005; Zhang et al., 2018; Zu et al., 2017), 8 used O₃-1 h max, 28 used O₃-8 h max, 7 used O₃-24 h avg, 1 study reported both O₃-1 h max and O₃-24 h avg, while 3 studies reported risk estimates based on O₃-1 h max and O₃-8 h max. Among these studies, 16 were carried out in North America, 15 in Europe, 16 were conducted in other regions. Across these studies, 17 specified RRs were observed during the warm season, 11 took the cold season into consideration. These studies estimated RRs for children (30 studies), adults (19 studies) and the elderly (11 studies). The quality for most included studies was moderate to high. The characteristics of these studies are summarized in Table 1, along with the scientific evidence reporting the presence or the absence of a significant association between asthma exacerbations and ozone exposure with different average times (Table 2).

3.1. Overall analyses

The association between ozone and asthma risk was statistically significant for both O₃-1 h max (RR, 1.012; 95%CI, 1.005–1.019) and O₃-8 h max (1.011; 1.007–1.014) but was marginally significant for O₃-24 h avg (1.005; 0.996–1.014). Table 3 presents specific RR and 95%CI before and after standardization for each original study.

The main results of the meta-analysis are presented in Table 4. Forest plots for O₃-1 h max, O₃-8 h max and O₃-24 h avg are presented in Figs. 2–4, respectively. The funnel plots were symmetric (Supplementary Figs. 1–3). Substantial heterogeneity was observed for all the three ozone averaging times (*I*² = 75.06%, 90.56% and 88.35%, respectively). But no significant publication bias was observed using Egger's test (*P* = 0.27, 0.17 and 0.95, respectively).

3.2. Subgroup analyses

During the warm season, ozone was significantly associated with asthma exacerbations. O₃-1 h max had the highest RR of 1.014 (95%CI, 1.005–1.024), followed by O₃-8 h max (RR, 1.012; 95%CI, 1.009–1.016), while marginal association was identified for O₃-24 h avg (RR, 1.008; 95%CI, 0.998–1.017). During the cold season, null associations were observed for all the three averaging times.

Taking region into consideration, positive associations were observed in North America, with O₃-1 h max (RR, 1.013; 95%CI, 1.009–1.017), O₃-8 h max (RR, 1.012; 95%CI, 1.008–1.015) and O₃-24 h avg (RR, 1.013; 95%CI, 1.009–1.017). However, no significant associations were found in Europe.

In children, a 10 µg/m³ increase of ozone was associated with RRs of 1.015 (95%CI: 1.001–1.017) and 1.011 (95%CI: 1.004–1.019) in asthma exacerbations for O₃-1 h max and O₃-8 h max, respectively, but not for O₃-24 h avg (RR, 1.000; 95%CI, 0.985–1.015). In adults, the estimates were comparable for O₃-24 h avg (RR, 1.014; 95% CI, 1.007–1.021) and O₃-8 h max (RR, 1.012; 95%CI, 1.006–1.018), but not significant for O₃-1 h max (RR, 1.010; 95% CI, 0.993–1.027). In the elderly subgroup, we found positive associations for O₃-1 h max (RR, 1.015; 95%CI, 1.007–1.023), O₃-8 h max (RR, 1.007; 95%CI, 1.001–1.013), and O₃-24 h avg (RR, 1.022; 95%CI, 1.014–1.031).

For most of the subgroups we didn't identify significant publication bias (*P* > 0.05), except O₃-8 h max during the warm season (*P* = 0.01).

High heterogeneity (*I*² > 50) was found in different seasons except O₃-1 h max during the cold season and O₃-24 h avg during the warm season. Studies conducted in different regions using O₃-1 h max and O₃-24 h avg were not significantly heterogeneous (*I*² ≤ 50). However,

Table 1
Characteristics of included studies.

Study	Country	Study period	Population	Asthma exacerbation outcome	Diagnosis standard	Ozone averaging time	Study quality score
Castellsague J, 1995	Spain	1985–1989	C, A, E	EMV	Standards developed by a panel of experts	1 h	3
Schouten JP, 1996	Netherlands	1977–1989	C, A, E	HA	ICD-9: 493	8 h, 1 h	4
Stieb DM, 1996	Canada	1984–1992	C, A, E	EMV	NA	1 h, 24 h	2
Medina S, 1997	France	1991–1995	C, A	HA	SOS-Medecins R01	8 h	3
Sunyer J, 1997	Spain, Finland, France, England	1986–1992	C, A	EMV	ICD-9: 493	8 h, 1 h	3
Anderson HR, 1998	England	1987–1992	C, A, E	HA	ICD-9: 493	8 h	5
Morgan G, 1998	Australia	1990–1994	C, A	HA	ICD-9: 493	1 h	5
Tenias JM, 1998	Spain	1993–1995	A, E	EMV	A list of symptoms of asthma	1 h	3
Cassino C, 1999	USA	1992–1995	A, E	EMV	ICD-9: 493–493.9	24 h	4
Hajat S, 1999	England	1992–1994	C, A, E	EMV	ICD-9: 493	8 h	5
Sheppard L, 1999	USA	1987–1994	C, A	HA	ICD-9: 493	8 h	4
Wong TW, 1999	China	1994–1995	C, A, E	HA	ICD-9: 493	8 h	5
Gouveia N, 2000	Brazil	1992–1994	C	HA	ICD-9: 493	1 h	5
Hernandez-Cadena L, 2000	Mexico	1998–1999	C, A, E	EMV	ICD-9: 493	1 h	4
Fusco D, 2001	Italy	1995–1997	C	HA	ICD-9: 493	8 h	5
Petroeschevsky A, 2001	Australia	1987–1994	C, A	HA	ICD-9: 493	8 h	5
Thompson AJ, 2001	Northern Ireland	1993–1995	C, A, E	EMV	NA	24 h	2
Lee JT, 2002	Korea	1997–1999	C	HA	ICD-10: J45-J46	1 h	4
Wong CM, 2002	China, England	1995–1997, 1992–2004	A	EMV	ICD-9: 493	8 h	5
Galan I, 2003	Spain	1995–1998	C, A, E	EMV	ICD-9: 493	8 h	5
Jaffe DH, 2003	USA	1991–1996	C, A, E	EMV	ICD-9: 493	8 h	4
Lin M, 2004	Canada	1987–1998	C	HA	ICD-9: 493	1 h	4
Wilson AM, 2004	England	1996–2000	C, A, E	EMV	ICD-9: 493	8 h	5
Lee JT, 2006	Korea	2002	C	HA	ICD-10: J45-J46	8 h	4
Lee SL, 2006	China	1997–2002	C	HA	ICD-9: J49	8 h	5
Ito K, 2007	USA	1999–2002	C, A, E	EMV	ICD-9: 493	8 h	3
Ko FWS, 2007	China	2000–2005	C, A, E	HA	ICD-9: 493	8 h	5
Andersen ZJ, 2008	Denmark	2001–2004	C	HA	ICD-9: 493	24 h	4
Babin S, 2008	USA	1994–2005	C, A	EMV	ICD-9: 493	8 h	3
Bell ML, 2008	China	1995–2002	C, A, E	HA	ICD-9: 493	8 h	5
Mohr LB, 2008	USA	2001–2003	C	EMV	ICD-9: 493	8 h	3
Szyszkowicz M, 2008	Canada	1992–2002	C	EMV	ICD-9: 493	24 h	4
Mar TF, 2009	USA	1998–2002	C, A	EMV	ICD-9: 493–493.9	8 h, 1 h	3
Stieb DM, 2009	Canada	1992–2003	C, A, E	EMV	ICD-9/10493:J45	8 h	5
Neidell M, 2010	USA	1989–1997	C, A, E	HA	ICD-9: 493	1 h	4
Samoli E, 2010	Athens	2001–2004	C	HA	ICD-9: 493	8 h	5
Silverman RA, 2010	USA	1999–2006	C, A, E	HA	ICD-9: 493	8 h	4
Amancio CT, 2012	Brazil	2004–2005	C	HA	ICD10: J45	8 h	4
Cirera L, 2012	Spain	1995–1998	C, A, E	EMV	By two trained nurses	8 h	2
Chakraborty P, 2013	India	2008–2010	C	EMV	ICD-9: 493/ICD-10: J45/4d	24 h	2
Son JY, 2013	Korea	2003–2008	C, A, E	HA	ICD-10: J45-J46	8 h	4
Chen K, 2016	Australia	2003–2013	C, A, E	HA	ICD-10: J45-J46	1 h	4
Goodman JE, 2017	USA	1999–2009	C, A, E	HA	ICD-9: 493	8 h	5
Zu K, 2017	USA	2001–2013	C, A, E	HA	ICD-9: 493	8 h	4
Zhang H, 2018	China	2013–2015	C, A, E	EMV, HA	NA	8 h	3
Dai Y, 2018	China	2005–2014	C, A, E	EMV	ICD-9: 493	24 h	5
Strosnider HM, 2018	USA	2001–2014	C, A, E	EMV	ICD-9: 493	8 h	5

Abbreviations: A, adult; C, child; E, elderly; ERV, Emergency room visit; HA, Hospital admission; NA, not available; 1 h, daily 1 h maximum; 8 h, daily maximum 8 h average; 24 h, daily average concentration.

we observed significant heterogeneity in different age groups including children (O_3 -1 h max, O_3 -8 h max, and O_3 -24 h avg), adults (O_3 -1 h max and O_3 -8 h max) and the elderly (O_3 -8 h max).

Results of subgroup analyses are presented in Table 4.

3.3. Meta-regression

We observed modification of the association between ozone and asthma exacerbations by daily average temperature. Higher ozone

Table 2
Scientific evidence on associations of ozone exposure and asthma exacerbations with different averaging time.

Averaging time	Statistically significant results		Lack of statistically significant evidence	
	No. of studies	Reference no(s).	No. of studies	Reference no(s).
O_3 -1 h max	6	15, 19, 26, 30, 32, 43	6	5, 23, 27, 37, 39, 45
O_3 -8 h max	17	7, 9, 10, 11, 18, 20, 21, 22, 25, 30, 31, 33, 34, 36, 41, 47, 48	14	6, 12, 13, 17, 23, 24, 28, 29, 35, 38, 40, 42, 49, 51
O_3 -24 h avg	3	8,14,16	5	17, 19, 44, 46, 50

Table 3

Summarized estimates of ambient ozone exposure with different averaging time and asthma exacerbations risk in original studies.

Study	Ozone averaging time	Season	Population	Lag day	Ozone concentration ($\mu\text{g}/\text{m}^3$)	Daily average temperature ($^{\circ}\text{C}$)	RR (95%CI) before standardization*	Standardized RR (95%CI)*
Castellsague J, 1995	1 h	Cold	C, A, E	2	42.45	NA	1.009 (1.000–1.018)	1.012 (0.999–1.025)
Castellsague J, 1995	1 h	Warm	C, A, E	2	63.23	NA	0.998 (0.990–1.007)	0.998 (0.987–1.010)
Schouten JP, 1996	8 h	Cold	C, A, E	1	53.00	4.56	1.019 (0.996–1.048)	1.019 (0.996–1.048)
Schouten JP, 1996	8 h	Warm	C, A, E	1	86.00	14.35	1.001 (0.999–1.003)	1.001 (0.999–1.003)
Schouten JP, 1996	1 h	Cold	C, A, E	1	46.50	4.56	1.015 (0.993–1.043)	1.021 (0.991–1.057)
Schouten JP, 1996	1 h	Warm	C, A, E	1	72.75	14.35	1.000 (0.983–1.019)	0.999 (0.978–1.025)
Stieb DM, 1996	1 h	Warm	C, A, E	2	61.15	19.40	1.018 (1.000–1.037)	1.024 (1.000–1.049)
Stieb DM, 1996	24 h	Warm	C, A, E	2	152.88	19.40	1.026 (0.998–1.055)	1.014 (0.999–1.029)
Medina S, 1997	8 h	Whole	C	0	34.00	12.00	1.032 (0.982–1.113)	1.032 (0.982–1.113)
Medina S, 1997	8 h	Whole	A	1	34.00	12.00	1.021 (0.999–1.048)	1.021 (0.999–1.048)
Anderson HR, 1998	8 h	Warm	C	0	30.38 ± 21.36^a	12.30^a	1.015 (0.995–1.035)	1.008 (0.997–1.018)
Anderson HR, 1998	8 h	Cold	C	0	30.38 ± 21.36^a	12.30^a	0.971 (0.942–1.000)	0.985 (0.970–1.000)
Anderson HR, 1998	8 h	Warm	A	1	30.38 ± 21.36^a	12.30^a	1.043 (1.018–1.067)	1.022 (1.009–1.034)
Anderson HR, 1998	8 h	Cold	A	1	30.38 ± 21.36^a	12.30^a	1.030 (0.994–1.067)	1.015 (0.999–1.034)
Anderson HR, 1998	8 h	Warm	E	1	30.38 ± 21.36^a	12.30^a	1.037 (0.989–1.087)	1.019 (0.994–1.044)
Anderson HR, 1998	8 h	Cold	E	2	30.38 ± 21.36^a	12.30^a	1.001 (0.931–1.077)	1.001 (0.965–1.039)
Morgan G, 1998	1 h	Whole	A	0	36.75 ± 19.11	17.80	0.995 (0.988–1.003)	0.994 (0.984–1.005)
Morgan G, 1998	1 h	Whole	C	1	36.75 ± 19.11	17.80	1.005 (0.995–1.014)	1.006 (0.994–1.019)
Tenias JM, 1998	1 h	Warm	A, E	1	55.5	23.00	1.083 (1.024–1.146)	1.111 (1.032–1.195)
Tenias JM, 1998	1 h	Cold	A, E	1	38.55	14.50	1.035 (0.966–1.11)	1.047 (0.955–1.147)
Cassino C, 1999	24 h	Whole	A	2	64.31	13.20	1.022 (0.998–1.054)	1.012 (0.999–1.029)
Hajat S, 1999	8 h	Whole	C	0	34.30 ± 22.54	11.90	0.983 (0.972–0.994)	0.983 (0.972–0.994)
Hajat S, 1999	8 h	Whole	A	0	34.30 ± 22.54	11.90	1.010 (1.000–1.019)	1.010 (1.000–1.019)
Hajat S, 1999	8 h	Whole	E	3	34.30 ± 22.54	11.90	0.983 (0.968–0.999)	0.983 (0.968–0.999)
Hajat S, 1999	8 h	Warm	C	1	44.49 ± 23.91	15.50	0.982 (0.969–0.996)	0.982 (0.969–0.996)
Hajat S, 1999	8 h	Warm	A	0	44.49 ± 23.91	15.50	1.015 (1.003–1.027)	1.015 (1.003–1.027)
Hajat S, 1999	8 h	Warm	E	0	44.49 ± 23.91	15.50	0.994 (0.976–1.014)	0.994 (0.976–1.014)
Hajat S, 1999	8 h	Cold	C	3	23.72 ± 14.90	8.30	0.985 (0.967–1.004)	0.985 (0.967–1.004)
Hajat S, 1999	8 h	Cold	A	0	23.72 ± 14.90	8.30	0.997 (0.980–1.014)	0.997 (0.980–1.014)
Hajat S, 1999	8 h	Cold	E	3	23.72 ± 14.90	8.30	0.970 (0.944–1.000)	0.970 (0.944–1.000)
Sheppard L, 1999	8 h	Whole	C, A	2	NA	11.78	1.015 (1.008–1.023)	1.015 (1.008–1.023)
Wong TW, 1999	8 h	Whole	C, A, E	0–2	$24.15(0-129.94)$	NA	1.031 (1.017–1.046)	1.031 (1.017–1.046)
Fletcher T, 2000	1 h	Whole	C	2	47.55 ± 28.58	19.40	1.001 (0.992–1.011)	1.001 (0.989–1.015)
Hernandez-Cadena L, 2000	1 h	Whole	C, A, E	1	75.86 ± 30.43	14.97	1.006 (0.995–1.016)	1.008 (0.993–1.021)
Fusco D, 2001	8 h	Whole	C	2	27.00 ± 17.30	16.80	1.023 (0.979–1.072)	1.023 (0.979–1.072)
Petroeschvsky A, 2001	8 h	Whole	C	1	$37.24(3.33-126.81)$	NA	1.033 (1.008–1.059)	1.033 (1.008–1.059)
Petroeschvsky A, 2001	8 h	Whole	A	2	$37.24(3.33-126.81)$	NA	1.043 (1.019–1.068)	1.043 (1.019–1.068)
Thompson AJ, 2001	24 h	Warm	C, A, E	0–1	68.72 ± 27.2	13.00	0.995 (0.944–1.051)	0.997 (0.970–1.027)
Thompson AJ, 2001	24 h	Cold	C, A, E	0–1	62.84 ± 28.67	6.40	0.944 (0.908–0.985)	0.970 (0.951–0.992)
Lee JT, 2002	1 h	Whole	C	1	52.92 ± 27.34	12.90	1.028 (1.016–1.038)	1.038 (1.022–1.050)
Sunyer J, 1997	8 h	Warm	C	1/2	$43.75(1.00-283.00)$	NA	1.030 (0.920–1.170)	1.008 (0.979–1.045)
Sunyer J, 1997	8 h	Cold	C	1/2	$32.81(0.75-212.25)$	NA	1.080 (0.960–1.200)	1.021 (0.989–1.053)
Sunyer J, 1997	1 h	Warm	A	0/1	$43.75(1.00-283.00)$	NA	0.990 (0.956–1.078)	0.997 (0.988–1.021)
Sunyer J, 1997	1 h	Cold	A	0/1	$32.81(0.75-212.25)$	NA	0.970 (0.916–1.032)	0.992 (0.978–1.009)
Galan I, 2003	8 h	Whole	C, A, E	1	43.75 ± 28.20	15.00	1.045 (1.018–1.073)	1.045 (1.018–1.073)
Jaffe DH, 2003	8 h	Whole	C, A, E	3	117.60 ± 39.20	NA	1.026 (1.000–1.056)	1.026 (1.000–1.056)
Jaffe DH, 2003	8 h	Whole	C, A, E	3	98 ± 33.32	NA	1.005 (0.985–1.026)	1.005 (0.985–1.026)
Jaffe DH, 2003	8 h	Whole	C, A, E	3	111.72 ± 31.36	NA	1.026 (0.995–1.056)	1.026 (0.995–1.056)
Lin M, 2004	1 h	Whole	C	0–5	41.19 ± 16.96	NA	0.969 (0.990–1.096)	1.083 (0.986–1.207)
Lin M, 2004	1 h	Whole	C	0–4	41.19 ± 16.96	NA	1.038 (0.990–1.096)	1.087 (0.995–1.198)
Lin M, 2004	1 h	Whole	C	0–5	41.19 ± 16.96	NA	0.976 (0.941–1.014)	0.968 (0.904–1.041)
Lin M, 2004	1 h	Whole	C	0–1	41.19 ± 16.96	NA	0.948 (0.917–0.979)	0.936 (0.894–0.986)
Wilson AM, 2004	8 h	Whole	C	1	84.48 ± 26.46	NA	1.010 (0.995–1.026)	1.010 (0.995–1.026)
Wilson AM, 2004	8 h	Whole	A	1	84.48 ± 26.46	NA	1.000 (0.990–1.010)	1.000 (0.990–1.010)
Wilson AM, 2004	8 h	Whole	E	1	84.48 ± 26.46	NA	0.995 (0.974–1.015)	0.995 (0.974–1.015)
Wilson AM, 2004	8 h	Whole	C	1	NA	NA	1.005 (0.974–1.036)	1.005 (0.974–1.036)
Wilson AM, 2004	8 h	Whole	A	1	NA	NA	0.995 (0.980–1.010)	0.995 (0.980–1.010)
Lee JT, 2006	8 h	Whole	C	3	58.47 ± 33.07	12.90	1.028 (1.009–1.047)	1.028 (1.009–1.047)
Lee SL, 2006	8 h	Whole	C	2	28.60 ± 16.00	23.70	1.026 (1.018–1.034)	1.026 (1.018–1.034)
Ito K, 2007	8 h	Warm	C, A, E	0–1	59.58 ± 37.24	NA	1.048 (1.026–1.073)	1.048 (1.026–1.073)
Ko FWS, 2007	8 h	Whole	C	0–5	43.40 ± 23.70	23.40	1.039 (1.030–1.048)	1.039 (1.030–1.048)
Ko FWS, 2007	8 h	Whole	A	0–5	43.40 ± 23.70	23.40	1.041 (1.032–1.050)	1.041 (1.032–1.050)
Ko FWS, 2007	8 h	Whole	E	0–5	43.40 ± 23.70	23.40	1.023 (1.015–1.030)	1.023 (1.015–1.030)
Andersen ZJ, 2008	24 h	Whole	C	0–5	88.20 ± 36.75	10.00	1.047 (0.906–1.247)	1.025 (0.950–1.132)
Babin S, 2008	8 h	Whole	C	NA	NA	NA	1.012 (1.001–1.023)	1.012 (1.001–1.023)
Babin S, 2008	8 h	Whole	A	NA	NA	NA	1.006 (1.000–1.012)	1.006 (1.000–1.012)
Mohr LB, 2008	8 h	Warm	C	1	115.64	NA	0.980 (0.949–1.015)	0.980 (0.949–1.015)
Mohr LB, 2008	8 h	Cold	C	1	39.20	NA	1.026 (0.985–1.071)	1.026 (0.985–1.071)
Szyszkowicz M, 2008	24 h	Warm	C (0–10 yrs)	1	68.36 ± 34.18^a	3.90^a	1.108 (1.041–1.18)	1.021 (1.008–1.035)
Szyszkowicz M, 2008	24 h	Whole	C (10 + yrs)	1	68.36 ± 34.18	3.90	1.084 (1.04–1.129)	1.016 (1.008–1.025)

Table 3 (continued)

Study	Ozone averaging time	Season	Population	Lag day	Ozone concentration ($\mu\text{g}/\text{m}^3$)	Daily average temperature ($^{\circ}\text{C}$)	RR (95%CI) before standardization*	Standardized RR (95%CI)*
Szyszkowicz M, 2008	24 h	Cold	A, E C (0–10 yrs)	1	68.36 $\pm$ 34.18 ^a	3.90 ^a	1.101 (1.041–1.163)	1.020 (1.008–1.032)
Szyszkowicz M, 2008	24 h	Warm	C (10 + yrs),	1	68.36 $\pm$ 34.18 ^a	3.90 ^a	1.063 (1.022–1.105)	1.012 (1.004–1.020)
Szyszkowicz M, 2008	24 h	Whole	A, E C (10 + yrs),	1	68.36 $\pm$ 34.18	3.90	1.040 (1.013–1.068)	1.008 (1.003–1.013)
Mar TF, 2009	8 h	Warm	C	3	76.83	15.00	1.056 (1.010–1.107)	1.056 (1.010–1.107)
Mar TF, 2009	1 h	Warm	C	3	57.62	15.00	1.041 (1.000–1.092)	1.054 (1.000–1.122)
Mar TF, 2009	8 h	Warm	A	5	76.83	15.00	1.056 (1.010–1.107)	1.056 (1.010–1.107)
Mar TF, 2009	1 h	Warm	A	3	57.62	15.00	1.041 (1.010–1.071)	1.054 (1.014–1.095)
Stieb DM, 2009	8 h	Whole	C, A, E	2	36.06	NA	1.009 (1.001–1.017)	1.009 (1.001–1.017)
Neidell M, 2010	1 h	Warm	C	0–4	120.54	NA	1.008 (1.003–1.014)	1.011 (1.003–1.018)
Neidell M, 2010	1 h	Warm	A	0–4	120.54	NA	1.009 (1.006–1.013)	1.012 (1.007–1.018)
Neidell M, 2010	1 h	Warm	E	0–4	120.54	NA	1.011 (1.006–1.017)	1.015 (1.007–1.023)
Samoli E, 2010	8 h	Whole	C	0	70.90	18.50	0.969 (0.935–1.005)	0.969 (0.935–1.005)
Silverman RA, 2010	8 h	Warm	C	0–1	81.50	NA	1.021 (1.009–1.035)	1.021 (1.009–1.035)
Silverman RA, 2010	8 h	Warm	C	0–1	81.50	NA	1.046 (1.026–1.067)	1.046 (1.026–1.067)
Silverman RA, 2010	8 h	Warm	A	0–1	81.50	NA	1.019 (1.009–1.028)	1.019 (1.009–1.028)
Silverman RA, 2010	8 h	Warm	E	0–1	81.50	NA	1.014 (1.005–1.023)	1.014 (1.005–1.023)
Amancio CT, 2012	8 h	Whole	C	1	74.30 $\pm$ 32.40	NA	0.968 (0.997–1.028)	0.968 (0.997–1.028)
Cirera L, 2012	8 h	Whole	C, A, E	0	80.00 $\pm$ 18.00	20.00	0.971 (0.938–1.006)	0.971 (0.938–1.006)
Chakraborty P, 2013	24 h	Whole	C	NA	NA	NA	0.988 (0.978–0.999)	0.994 (0.988–0.999)
Son JY, 2013	8 h	Warm	C, A, E	1	65.55 ^a	14.11 ^a	1.073 (1.029–1.119)	1.019 (1.008–1.031)
Son JY, 2013	8 h	Cold	C, A, E	1	65.55 ^a	14.11 ^a	1.062 (0.959–1.176)	1.016 (0.989–1.046)
Chen K, 2016	1 h	Whole	C	0–4	28.52 $\pm$ 11.64	17.29	1.036 (1.009–1.064)	1.048 (1.012–1.085)
Chen K, 2016	1 h	Whole	A	0–4	28.52 $\pm$ 11.64	17.29	1.034 (1.004–1.066)	1.046 (1.005–1.088)
Chen K, 2016	1 h	Warm	C, A	0–4	NA	NA	1.100 (1.053–1.148)	1.068 (1.036–1.101)
Chen K, 2016	1 h	Cold	C, A	0–4	NA	NA	0.98 (0.912–1.053)	0.986 (0.940–1.036)
Goodman JE, 2017	8 h	Whole	C (0–6 yrs)	0–1	60.23 $\pm$ 33.12	13.21	1.000 (0.993–1.007)	1.000 (0.993–1.007)
Goodman JE, 2017	8 h	Whole	C (6–18 yrs)	0–1	60.23 $\pm$ 33.12	13.21	1.010 (1.001–1.019)	1.010 (1.001–1.019)
Goodman JE, 2017	8 h	Whole	A	0–1	60.23 $\pm$ 33.12	13.21	1.006 (1.000–1.012)	1.006 (1.000–1.012)
Goodman JE, 2017	8 h	Whole	E	0–1	60.23 $\pm$ 33.12	13.21	1.000 (0.994–1.005)	1.000 (0.994–1.005)
Goodman JE, 2017	8 h	Warm	C, A, E	0–1	NA	NA	1.003 (0.998–1.009)	1.003 (0.998–1.009)
Zu K, 2017	8 h	Whole	C	1	63.11 $\pm$ 23.52	19.83	1.017 (1.010–1.024)	1.017 (1.010–1.024)
Zu K, 2017	8 h	Whole	A	1	63.11 $\pm$ 23.52	19.83	1.006 (1.002–1.011)	1.006 (1.002–1.011)
Zu K, 2017	8 h	Whole	E	1	63.11 $\pm$ 23.52	19.83	1.006 (0.998–1.013)	1.006 (0.998–1.013)
Zhang H (EMV), 2018	8 h	Whole	C, A, E	0–3	100.00 $\pm$ 44.00	17.00	1.003 (0.990–1.016)	1.003 (0.990–1.016)
Zhang H (HA), 2018	8 h	Whole	C, A, E	0–3	100.00 $\pm$ 44.00	17.00	1.033 (1.003–1.067)	1.033 (1.003–1.067)
Dai Y, 2018	24 h	Whole	C	0–6	NA	NA	0.979 (0.966–0.993)	0.989 (0.982–0.996)
Dai Y, 2018	24 h	Whole	A	0–6	NA	NA	1.027 (1.013–1.042)	1.014 (1.007–1.022)
Dai Y, 2018	24 h	Whole	E	0–6	NA	NA	1.042 (1.027–1.058)	1.022 (1.014–1.031)
Strosnider HM, 2018	8 h	Whole	C	0–6	NA	NA	1.011 (1.008–1.014)	1.011 (1.008–1.014)
Strosnider HM, 2018	8 h	Whole	A	0–6	NA	NA	1.018 (1.015–1.020)	1.018 (1.015–1.020)
Strosnider HM, 2018	8 h	Whole	E	0–6	NA	NA	1.005 (0.999–1.012)	1.005 (0.999–1.012)
Wong CM, 2002	8 h	Whole	A	2	33.50 $\pm$ 23.00	23.20	1.012 (1.000–1.024)	1.012 (1.000–1.024)
Wong CM, 2002	8 h	Whole	A	0	34.90 $\pm$ 23.10	11.90	0.993 (0.983–1.003)	0.993 (0.983–1.003)
Bell ML, 2008	8 h	Whole	C, A, E	1	41.94	23.10	1.021 (1.006–1.037)	1.021 (1.006–1.037)

Abbreviations: A, adult; C, child; E, elderly; ERV, Emergency room visit; HA, Hospital admission; NA, not available; 1 h, daily O₃-1 h maximum; 8 h, daily maximum O₃-8 h average; 24 h, daily O₃ average concentration.

The Ozone concentration was presented as (mean $\pm$ SD) or mean (range) with available data.

RR (95%CI) before standardization*: RR (95%CI) presented here are estimates corresponding to a 10 $\mu\text{g}/\text{m}^3$ increase of ozone in different averaging time estimates before transferred into O₃-8 h max.

Standardized RR (95%CI)*: the RR and 95%CI in other averaging time (O₃-1 h max and O₃-24 h avg) are transferred into RRs and 95%CI per 10 $\mu\text{g}/\text{m}^3$ increase of O₃-8 h max, by using a ratio of 20:15:8 for O₃-1 h max: O₃-8 h max: O₃-24 h avg.

^a Only ozone concentration and temperature of the whole year are available.

effect estimates were identified in cities with higher daily average temperatures ($P = 0.033$).

3.4. Sensitivity analyses

When study quality was taken into consideration, it did not alter the overall results across the ozone averaging times. The magnitude RR was highest for O₃-1 h max (RR, 1.010; 95%CI, 1.003–1.016; $I^2 = 66.00\%$; Egger's test, $P = 0.69$), while no significant association was found for O₃-24 h avg (Table 5).

In sensitivity analyses restricting to studies using ≤ 3 lag days or using RRs before standardization, we observed similar patterns with

the smallest magnitude RR for O₃-24 h avg. The results are presented in Table 5.

4. Discussion

In this meta-analysis, we evaluated the risks of asthma exacerbations associated with short-term ozone exposure when three different averaging times for ambient ozone concentration (O₃-1 h max, O₃-8 h max and O₃-24 h avg) are used. Overall, we found that risk estimates using O₃-1 h max and O₃-8 h max were significant and similar, and both were quite dissimilar to the effect based on O₃-24 h avg, which was only marginally significant. The difference was robust in subgroup

Table 4

Summarized results of overall and subgroup analyses on ambient ozone exposure with different averaging time and asthma exacerbations risk.

	O ₃ -1 h max			O ₃ -8 h max			O ₃ -24 h avg		
	RR(95%CI)#	I ² (%)	Egger's test, P	RR(95%CI)#	I ² (%)	Egger's test, P	RR(95%CI)#	I ² (%)	Egger's test, P
Overall	1.012 (1.005–1.019)	75.06	0.27	1.011 (1.007–1.014)	90.56	0.17	1.005 (0.996–1.014)	88.35	0.95
Season									
Warm	1.014 (1.005–1.024)	87.23	0.14	1.012 (1.009–1.016)	88.89	0.01	1.008 (0.998–1.017)	0.00	0.46
Cold	1.007 (0.994–1.020)	30.41	0.67	0.999 (0.988–1.011)	52.62	0.34	0.995 (0.948–1.045)	93.39	NA
Region									
Europe	1.005 (0.997–1.013)	0.27	0.07	1.002 (0.996–1.007)	73.92	0.76	0.984 (0.959–1.010)	38.12	0.49
North America	1.013 (1.009–1.017)	5.42	0.54	1.012 (1.008–1.015)	81.42	0.62	1.013 (1.009–1.017)	18.94	0.07
Age									
Children	1.015 (1.001–1.017)	77.96	0.76	1.011 (1.004–1.019)	89.43	0.85	1.000 (0.985–1.015)	88.91	0.61
Adults	1.010 (0.993–1.027)	85.75	0.94	1.012 (1.006–1.018)	87.22	0.64	1.014 (1.007–1.021)	0.00	NA
Elderly	1.015 (1.007–1.023)	NA	NA	1.007 (1.001–1.013)	70.52	0.58	1.022 (1.014–1.031)	NA	NA

Abbreviations: NA, not available, due to the limited available studies in the subgroup; O₃-1 h max, daily 1 h maximum; O₃-8 h max, daily maximum 8 h average; O₃-24 h avg, daily average concentration.

RR(95%CI)#: the RR and 95%CI presented here are standardized values, which means RR(95%CI) in other averaging time (O₃-1 h max and O₃-24 h avg) are transferred into RR(95%CI) per 10 µg/m³ increase of O₃-8 h max, by using a ratio of 20:15:8 for O₃-1 h max: O₃-8 h max: O₃-24 h avg.

analysis and sensitivity analysis, and it was not altered by standardization of the RRs.

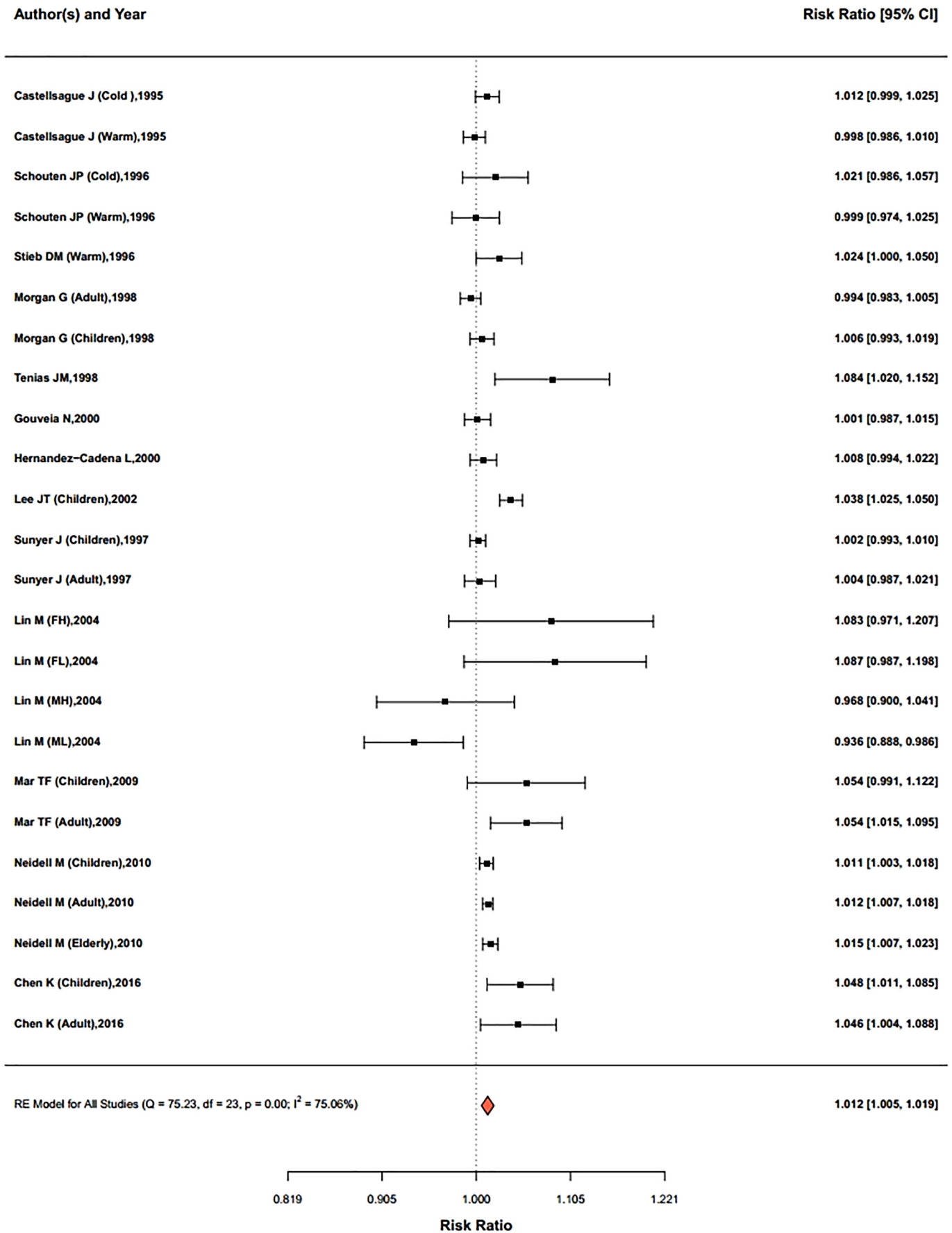
This finding is consistent with a case-crossover study showing that the respiratory effects of O₃-1 h max and O₃-8 h max were more comparable to each other, and larger than that of O₃-24 h avg (Darrow et al., 2011).

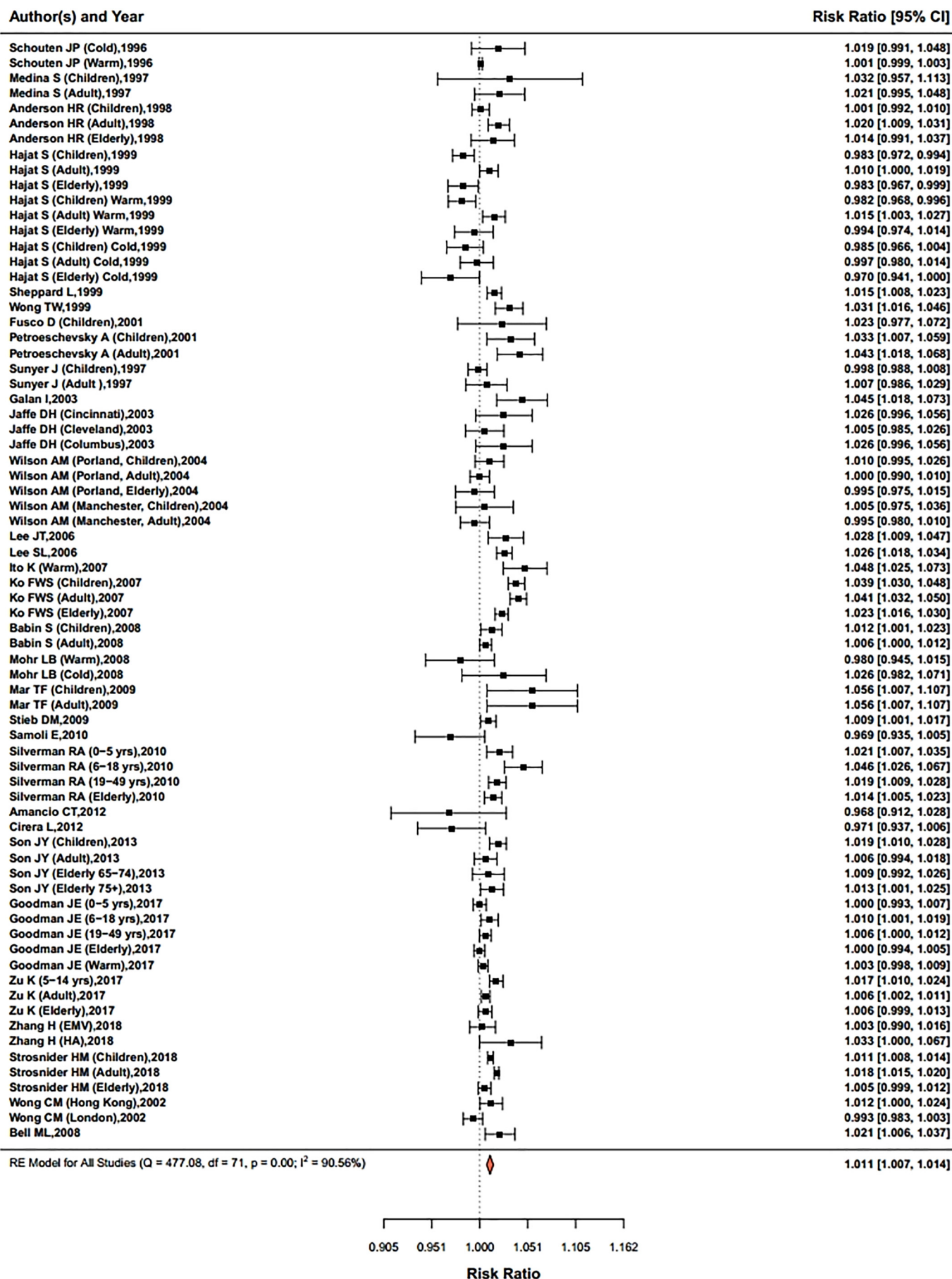
The biological mechanism may rely on the pathophysiology of asthma exacerbation (Papi et al., 2018). Known as acute or subacute episodes of increased symptoms, asthma exacerbations are characterized by progressively increasing shortness of breath, cough, wheezing or chest tightness, and decreasing lung function (Papi et al., 2018). An exacerbation can be triggered or promoted by acute high-dose stimulations (Darrow et al., 2011). Compared to O₃-24 h avg, O₃-1 h max and O₃-8 h max reflect peak pollution levels in a short time period rather than average levels of a day (Darrow et al., 2011). In addition, Lange (2018) found concentration of O₃-8 h max was well correlated with O₃-1 h max ($R^2 = 0.91$), but poorly correlated with the O₃-24 h avg value ($R^2 = 0.39$). Therefore, O₃-1 h max and O₃-8 h max may be more biologically relevant to acute asthma exacerbations than O₃-24 h avg. Recently, a panel study also found that ambient O₃-8 h max but not O₃-24 h avg. was associated with reduction in forced expiratory volume in 1 s (FEV₁) and peak expiratory flow (PEF) in chronic obstructive pulmonary disease (COPD) patients (Li et al., 2018). FEV₁ and PEF are indicators for hyper-responsiveness (Seltzer et al., 1986), which is one of the characteristic features of asthma (Guarnieri and Balmes, 2014).

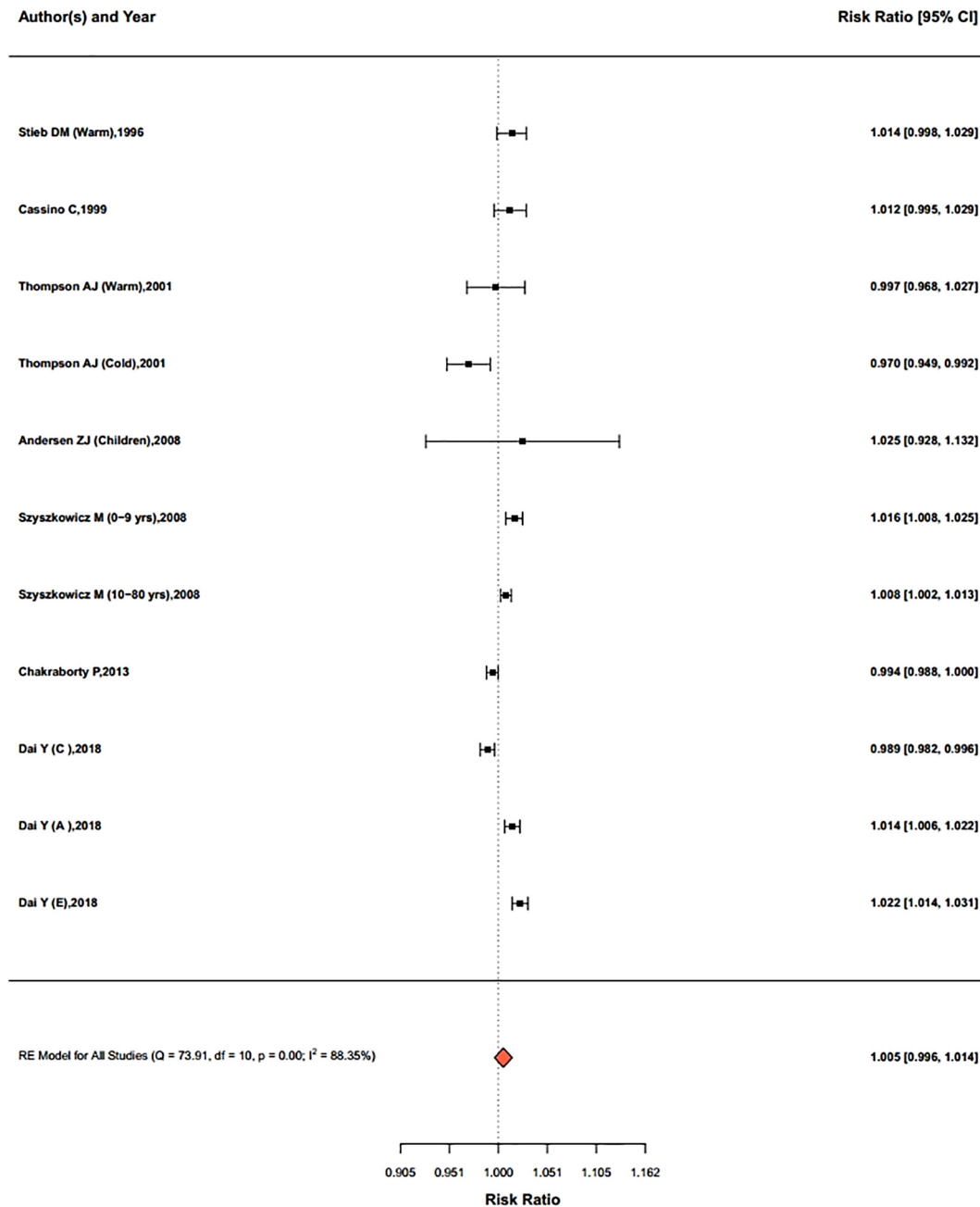
Significant heterogeneity was identified in this meta-analysis, which should be considered with caution. Several factors may contribute to the heterogeneity. Firstly, significant effects were observed during the warm season for O₃-1 h max, O₃-8 h max and O₃-24 h avg, but no effects were identified during the cold season. While temperature has long been known as an important driver of ozone episodes, we also found daily average temperature could increase the effect of ozone on asthma exacerbations. This may be one reason of seasonal difference. Another potential reason may be the exposure-relevant behavior during the warm *versus* the cold season. People tend to spend more time outside during the warm season (BL, 2011), which could increase their personal ozone exposure. Second, the association varied by regions. In contrast to significant positive findings in North America, we observed no significant associations of

ozone with asthma exacerbations in Europe. This heterogeneity might be partly due to differences in ozone concentrations. we noticed that the ozone concentrations were higher in North America (mean value was 71.51 µg/m³) than those in Europe (mean value was 48.97 µg/m³). Thirdly, our result showed the adverse effect in all age groups, which is different from the age-dependent pattern Dai et al. (2018) reported. However, due to limited number of studies using O₃-1 h max and O₃-24 h avg in the elderly groups, the result should be interpreted with caution. Another reason is related to the diagnosis of asthma. A range of factors could introduce spectrum-of-disease bias (Collaborators, 2017), including the difficulty of asthma diagnosis in children under 5 (2007), the common misclassification of asthma as COPD in the elderly (Yáñez et al., 2014), and an actual overlap of asthma and COPD (Burney et al., 2015) (asthma COPD overlap; ACO). Lastly, it is possible that some effect modifiers (e.g. sociodemographic data, pollen and smoke exposure), which were not available in the included articles, could have influenced these results. This may be the reason why the heterogeneity could not be completely controlled in our meta-analysis. However, with no detailed information of other asthma risk factors, we could not fully control the potential confounding. More efforts should be made to rule out substantial uncertainties and heterogeneities. It might be useful by applying new ozone exposure estimation methods in a fine scale (e.g. satellite-based aerosol optical depth models, land-use regression and chemical transport model simulation) (Di et al., 2016) to obtain more accurate results.

The relationship between ozone and asthma has been examined in several previous meta-analyses (Orellano et al., 2017; Zhang et al., 2016; Zheng et al., 2015), each finding a statistically significant association. However, to the best of our knowledge, this is the first meta-analysis to explore the associations between multiple ozone averaging times (O₃-1 h max, O₃-8 h max and O₃-24 h avg) and asthma exacerbations. Our findings indicate that using O₃-1 h max or O₃-8 h max of ambient concentrations represents short-term ozone exposure more relevant to asthma exacerbation risks than using 24-hour average concentration. Secondly, to avoid the influence of different ozone averaging time, we used a uniform ratio to convert ozone concentrations and RRs into the same metric, which made the effect estimates more comparable; and the results showed that the effects were not altered by

Fig. 2. Forest plot for the association between O₃-1 h max and asthma exacerbations.

Fig. 3. Forest plot for the association between O₃-8 h max and asthma exacerbations.

Fig. 4. Forest plot for the association between O₃-24 h avg and asthma exacerbations.

standardization. Moreover, even though high heterogeneity was observed, summary estimates were substantially unchanged in subgroup analysis and sensitivity analysis.

This study also has limitations. First, asthma exacerbations data were relied on diagnostic information provided by administrative data or hospital data record, several studies did not use ICD as diagnosis

Table 5

Summarized results in sensitivity analysis on ambient ozone exposure with different averaging time and asthma exacerbations risk.

	O ₃ -1 h max			O ₃ -8 h max			O ₃ -24 h avg		
	RR(95%CI)	I ² (%)	Egger's test, P	RR(95%CI)	I ² (%)	Egger's test, P	RR(95%CI)	I ² (%)	Egger's test, P
Study quality score ≥ 5	1.010 (1.003–1.016) ^b	66.00	0.69	1.008 (1.003–1.014) ^b	92.67	0.20	1.008 (0.989–1.028) ^b	94.82	0.17
Lag days ≤ 3	1.010 (1.001–1.019) ^b	75.10	0.27	1.010 (1.007–1.014) ^b	90.98	0.23	1.006 (0.997–1.016) ^b	86.84	0.80
Estimates before standardization ^a	1.009 (1.004–1.014) ^a	74.57	0.28	1.011 (1.007–1.014) ^a	90.56	0.17	1.009 (0.992–1.026) ^a	88.41	0.99

Abbreviations: O₃-1 h max, daily 1 h maximum; O₃-8 h max, daily maximum 8 h average; O₃-24 h avg, daily average concentration.

^a Estimates corresponding to a 10 µg/m³ increase of ozone in different averaging time estimates before transferred into O₃-8 h max.

^b Estimates corresponding to a 10 µg/m³ increase of O₃-8 h max, which are transferred by using a ratio of 20:15:8 for O₃-1 h max: O₃-8 h max: O₃-24 h avg.

standards (Boutin-Forzano et al., 2004; Castellsague et al., 1995; Cirera et al., 2012; Medina et al., 1997; Tenías et al., 1998). Thus, it can be potentially affected by the misclassification. But in sensitivity analysis, the result was robust after exclusion of these studies. Second, included studies used ambient ozone concentration from a small number of fixed outdoor monitors to represent the ozone exposure. It is likely that highly urbanized city centers will not only experience higher concentrations but also greater fluctuations in concentrations (Shah et al., 2015). Thus, the overall associations of ozone and asthma exacerbations may be underestimated in such ecological studies. Thirdly, there is a tendency that the positive results are more likely to be published. Therefore, it's important to clarify if the significant results are influenced by the publication bias, especially in studies with small but significant statistical effect magnitude. In addition, there may be a concern that using a constant ratio to convert among ozone averaging times adds bias and uncertainty, because the relationships among O_3 -1 h max, O_3 -8 h max and O_3 -24 h avg may have spatial and temporal variations (Anderson and Bell, 2010). However, ignoring the difference of ozone concentration averaging times and directly comparing values derived using different averaging times can lead to incorrect and misleading health risk conclusions (Lange, 2018). Given most original studies did not provide sufficient data to apply study-specific ratios, the best we can do in the present study was to use the most widely used conversion ratios. To solve this conversion problem in the future, we suggest that future ozone-related health effects studies present results from multiple ozone averaging times, if possible. Finally, due to the limited number of studies, we could not conduct subgroup analysis to evaluate gender and other potential modifiers.

5. Conclusion

The present meta-analysis confirmed the robustness about the association between ozone exposure and asthma exacerbations measured as ERVs and HAs. The association was significant during the warm season but not in the cold season. The association was more robust in regions where ambient ozone concentrations were higher. Our findings support the use of shorter averaging time of ambient ozone concentration within a day as a more biologically relevant measure of ozone exposure. Using either 1-hour or 8-hour daily maximum concentrations, when scaled to the same averaging time, yielded nearly identical estimates of relative risks of asthma exacerbation per unit increase in ozone exposure across all the studies included in this meta-analysis.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2019.06.382>.

Competing interests

The authors declare no competing interests.

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